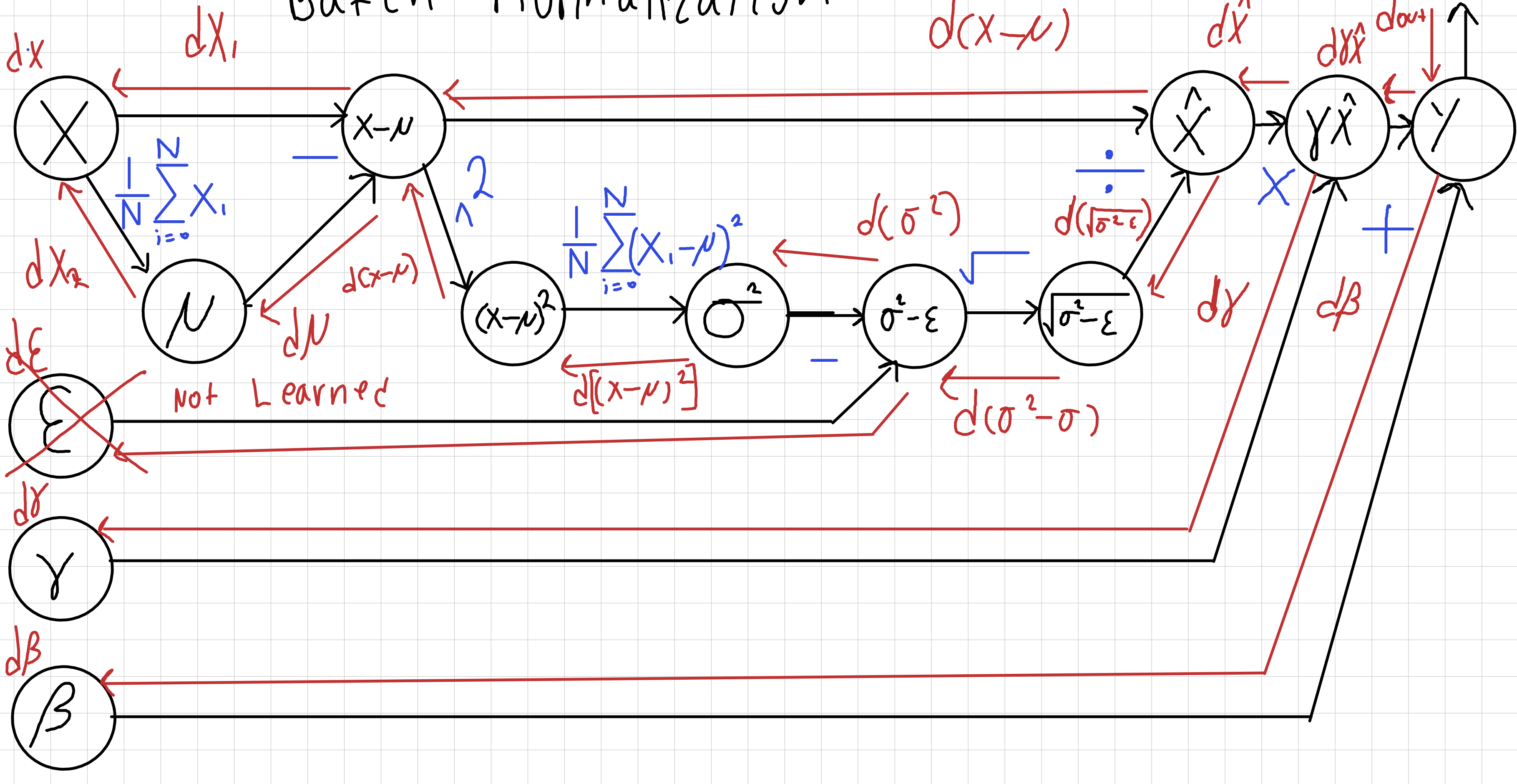
Batch normalization



$dout \in \mathbb{R}^{N\times D}$ $\quad \beta \in \mathbb{R}^{D}$ $\quad \gamma\hat{x} \in \mathbb{R}^{N\times D}$ $\quad \hat{x} \in \mathbb{R}^{N\times D}$

$\gamma \in \mathbb{R}^{D}$ $\quad \sqrt{\sigma^2-\varepsilon} \in \mathbb{R}^{D}$ $\quad (\sigma^2-\varepsilon) \in \mathbb{R}^{D}$ $\quad \sigma^2 \in \mathbb{R}^{D}$

$(x-\mu)^2 \in \mathbb{R}^{N\times D}$ $\quad (x-\mu) \in \mathbb{R}^{N\times D}$ $\quad \mu \in \mathbb{R}^{D}$ $\quad x \in \mathbb{R}^{N\times D}$

$$d\gamma\hat{x} = \frac{dy}{d\gamma\hat{x}} \cdot dout = \frac{d(\gamma\hat{x}+\beta)}{d\gamma\hat{x}} \cdot dout = dout$$

$$d\beta = \frac{dy}{d\beta} \cdot dout = \sum dout$$ because β is a constant but is added in a per element manner

$$d\hat{x} = \frac{d\gamma\hat{x}}{d\hat{x}} \cdot dy = \frac{d\gamma\hat{x}}{dx} \cdot d(\gamma\hat{x}+\beta) = \gamma \cdot d\gamma\hat{x} = \gamma\, dout$$

$$d\gamma = \frac{d\gamma\hat{x}}{d\gamma} \cdot dy = \hat{x}\, d\gamma\hat{x} = \hat{x}\, dout$$

$$d(\sqrt{\sigma^2+\varepsilon}) = d\hat{x} \cdot \frac{d([x-\mu]\{(\sigma^2-\varepsilon)^{1/2}\}^{-1})}{(\sigma^2-\varepsilon)^{1/2}} = d\hat{x} \cdot \left(-\frac{x-\mu}{(\sigma^2-\varepsilon)^{\cancel{1/2}\,2}}\right)$$

$$d(x-\mu) = d\hat{x} \cdot \frac{1}{\sqrt{\sigma^2-\varepsilon}}$$

$(\sigma^2+\varepsilon)^{1/2} \quad : 2 \cdot \frac{1}{2}\,($

$$d\sigma^2 = \frac{d(\sqrt{\sigma^2+\varepsilon})}{d\sigma^2} = d(\sqrt{\sigma^2+\varepsilon}) \cdot \frac{1}{2\sqrt{\sigma^2+\varepsilon}}$$

$$d([x-\mu]^2) = \frac{d\sigma^2}{d([x-\mu]^2)} = d\sigma^2 \cdot \frac{d\left(\frac{1}{N}\sum (x_i-\mu)^2\right)}{d([x_i-\mu]^2)} = d\sigma^2 \frac{1}{N} \begin{bmatrix} 1 \\ 1 \end{bmatrix}^{N\times D}$$

$$d(x-\mu)_1 = \frac{d([x-\mu]^2)}{d(x-\mu)} = \frac{d([x-\mu]^2)}{d(x-\mu)} = 2(x-\mu)\, d([x-\mu]^2)$$

$$d\mu = \frac{d(x-\mu)}{d\mu} = d(x-\mu) \cdot -1 = -1 \cdot (d(x-\mu)_1 + d(x-\mu)_2)$$

$$dx_1 = \frac{d(x-\mu)}{dx_1} = d(x-\mu)_1 + d(x-\mu)_2$$

$$dx_2 = \frac{1}{N}\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot d\mu$$

$$dx = dx_1 + dx$$

2.0 $$O[n,f,i,j] = \sum_{f=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} X[n, :, i \cdot s + h, j \cdot s + w] \cdot W[f, :, h, w] + b[f]$$

- filters applied to all channels therefore : in c index
- Per Pixel Gradient

2.1 $$\frac{\partial L[n,f,h,w]}{\partial W[f,:,h,w]} = \frac{\partial L_{pixel}}{\partial O_{pixel}} \cdot \frac{\partial O_{pixel}}{\partial W_{pixel}}$$

$$= \sum_{f=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} X[n, :, i \cdot s + h, j \cdot s + w] \cdot dout[n, f, h, w]$$

2.2 $$\frac{\partial L[n,f,h,w]}{\partial W[f,:,h,w]} = \frac{\partial L_{pixel}}{\partial O_{pixel}} \cdot \frac{\partial O_{pixel}}{\partial X_{pixel}}$$

$$= \sum_{f=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} W[f, :, h, w] \cdot dout[n, f, h, w]$$

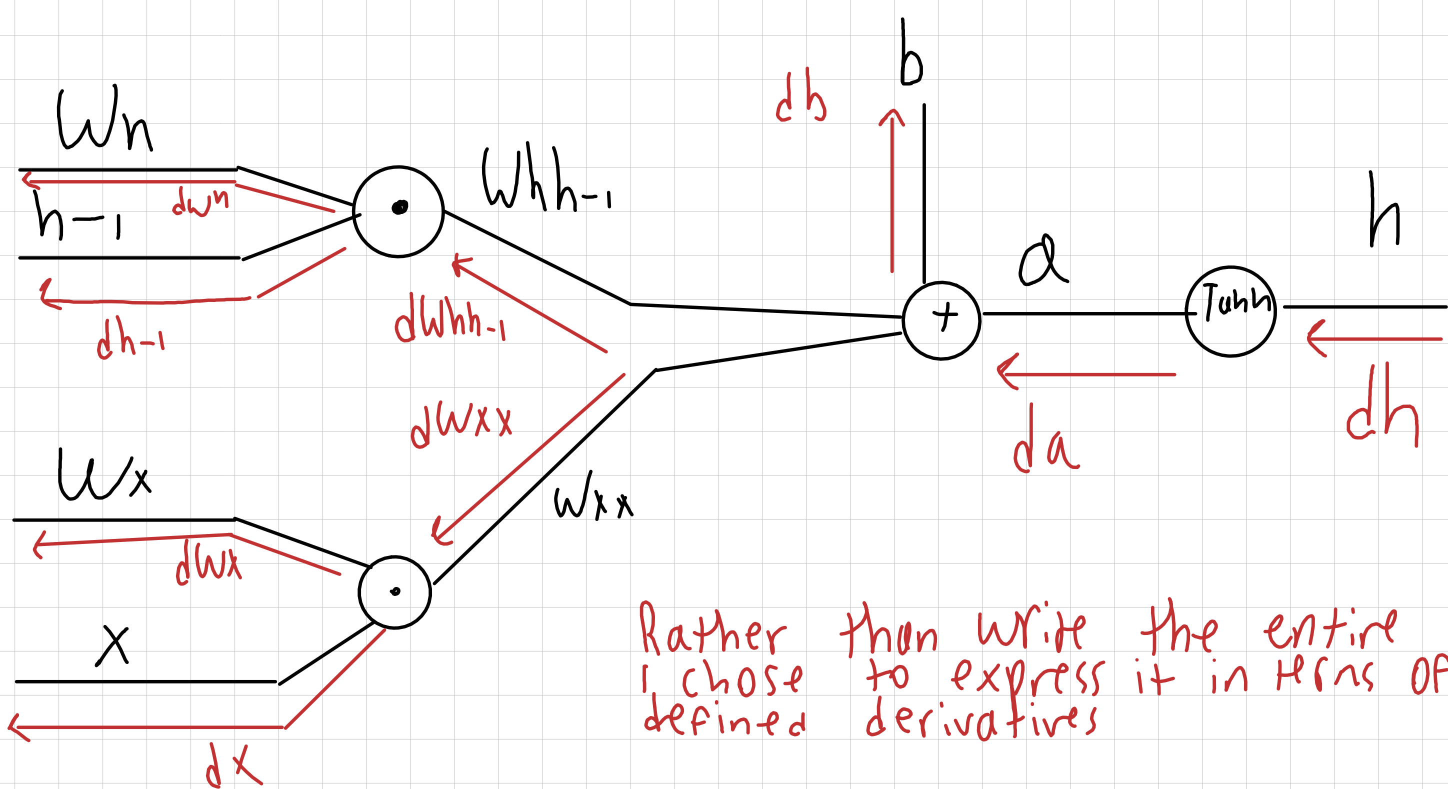
2.3 $$\frac{\partial L_{pixel}}{\partial b[f]} = \frac{\partial L_{pixel}}{\partial O_{pixel}} \cdot \frac{\partial O_{pixel}}{\partial X_{pixel}}$$

$$= \sum_{f=0}^{f-1} \sum_{h=0}^{h-1} \sum_{w=0}^{W-1} 1 \cdot dout[n, f, h, w$$

We pelform this accumulation becuuse, despite being a sing scalar per filter, the bias is appied to each element

Max pooling does not involve arithmetic in the same manner as the other problems here. Therefore backprop simply means applying the inputted gradient to the pixel that is used in the max pooling

RNN COMP Graph $h = \tanh(x \cdot W_x + h_{-1} \cdot W_h + b)$



Rather than write the entire derivative, I chose to express it in terms of previously defined derivatives

4.1 $da = \frac{d\tanh(a)}{da} dh = (1 - \tanh(a)^2)\, dh = (1 - h^2)\, dh$

4.2 $db = \frac{d(W_h h_{-1} + W_x x + b)}{db} \cdot da = \Sigma da$

b appears for each element so we need to sum

Useless computation

4.3 $dW_h h_{-1} = \frac{d(W_h h_{-1} + W_x X + b)}{d(W_h h_{-1})} da = 1 \cdot da$

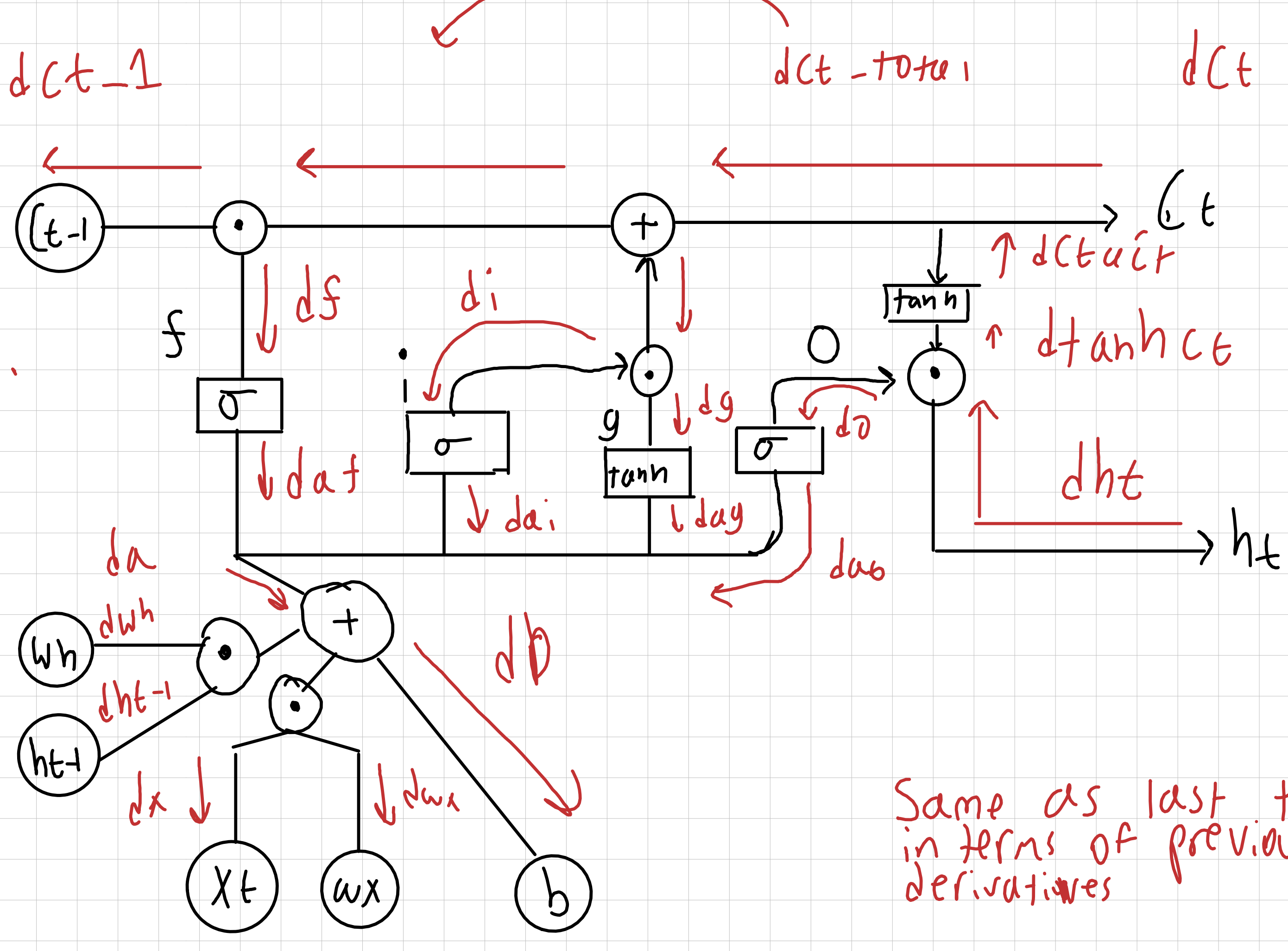
4.4 $dW_h = \frac{d(W_h h_{-1} + W_x X + b)}{d(W_h)} da = da \cdot (h_{t-1})^t$

4.5 $dh_{-1} = W_h^T da$

4.5 $dW_x = \frac{d(W_h h_{-1} + W_x X + b)}{d(W_x)} da = da X^T$

4.7 $dX = W_x^T da$

LSTM



Same as last time, express in terms of previously defined derivatives

$ht = tanh(ct)$

5.0 $dtanhCt = \frac{d[tanh(c(t)) \cdot O]}{dht^*} = O \cdot dht$

5.1 $dcact = \frac{dtanh(Ct)}{dCt} \cdot dtanhCt = dtanhct \cdot (1 - tanh(ct)^2)$

5.2 $dCt\_total = dct + dctact$

5.3 $dO = dht \cdot tanh(ct)$

5.4 $daO = dO \circ d\sigma(O) = dO \cdot (1-O) \cdot O$

5.5 $dg = dct\_total \cdot i$

5.6 $dag = dg \cdot dtanh(g) = dg \cdot (1 - g^2)$

5.7 $di = dCt\_total \cdot g$

5.8 $dai = di \cdot d\sigma(i) = di \cdot i \cdot (1-i)$

5.9 $df = dct\_total \cdot i$

5.10 $daf = df \cdot \sigma(f) = df \cdot f \cdot (1-f)$

5.11 $da = [dai; daf; dao; dag]$

5.12 $dx = \frac{d(WxX)}{dx} da = daWx^T$

5.13 $dWx = \frac{d(WxX)}{dwx} \cdot da = X^T \cdot da$

5.14 $dh_{-1} = \frac{d(Whht_{-1})}{d} \cdot da = daWh^T$

5.15 $dWh = \frac{d(Wht_{-1})}{dw} \cdot da = h_{t-1}^T \cdot da$

5.16 $db = \Sigma da$ because b is applied across each element